REAL ESTATE TRANSACTIONS ANALYSIS

Data Cleaning and Exploratory Data Analysis (EDA)

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1. Introduction

This project analyzes real estate transactions in Saudi Arabia using Python. The objectives are to inspect data quality, explore transaction patterns, examine geographical and temporal differences, calculate price per square meter, investigate extreme observations, and prepare a cleaned dataset for further analysis and visualization in Power BI.

2. Dataset Description

The dataset covers transactions from January 1, 2025, to December 31, 2025. It contains 201,556 records and 10 columns. The fields include region, city, city/neighborhood, transaction reference number, Gregorian and Hijri transaction dates, property type, number of properties, transaction price, and property area.

Dataset Description: The dataset contains the most important approved and registered real estate sale transactions in Saudi Arabia during 2025, recorded through electronic services, judicial entities, and private-sector notaries. It provides information on the number of transactions and properties, prices, and areas across regions, cities, and neighborhoods.
Publisher: Ministry of Justice, Saudi Arabia.
Source Platform: Saudi Open Data Platform.
URL: https://open.data.gov.sa/ar/home

Attribute	Description
المنطقة	Region
المدينة	City
المدينة /الحي	City / Neighborhood
الرقم المرجعي للصفقة	Transaction Reference Number
تاريخ الصفقة ميلادي	Gregorian Transaction Date
تاريخ الصفقة هجري	Hijri Transaction Date
تصنيف العقار	Property Type
عدد العقارات	Number of Properties
السعر	Transaction Price (SAR)
المساحة	Property Area (m^2)

3. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, and Power BI were used for data loading, inspection, preprocessing, statistical analysis, visualization, and dashboard development.

4. Data Inspection and Quality Assessment

4.1 Data Types and Date Conversion

The source dataset also contains a Hijri transaction date. The analysis uses the Gregorian date because the project examines transactions within the Gregorian calendar year 2025 and groups them by Gregorian month. The Hijri field was retained in the source data but was not used for the monthly analysis.

The Gregorian transaction date was initially stored as text. It was converted to datetime format using pandas.to_datetime() so that transactions could be grouped and analyzed by month. The converted dates ranged from 2025-01-01 to 2025-12-31, with zero missing dates after conversion.

4.2 Missing Values

The missing-value check returned zero missing values across all 10 columns. Therefore, no missing-value imputation was required.

4.3 Duplicate Records

The exact-row duplicate check returned zero duplicate rows. In addition, the transaction reference number was checked separately: all 201,556 records had a non-missing, unique reference number, with zero duplicate references.

4.4 Invalid Numeric Values

The checks found zero records with non-positive transaction prices or property areas. The number-of-properties field also contained no zero or negative values.

4.5 Dataset Quality Summary

Check	Result
Original records	201,556
Columns	10
Missing values	0
Exact duplicate rows	0
Non-positive price or area records	0
Date range	2025-01-01 to 2025-12-31

5. Descriptive Statistical Analysis

Descriptive statistics were used to understand the numerical variables. The large differences between the mean and median for transaction price and property area indicate highly dispersed distributions and motivated further examination of extreme values.

Variable	Mean	Median	Minimum	Maximum
Number of Properties	1.03	1	1	311
Transaction Price (SAR)	1,086,624	445,852.50	5,000	2,437,000,000
Property Area (m^2)	4,931.46	465.65	0.33	10,249,565

6. Investigation of Extreme Values

The notebook examined the highest transaction prices, largest property areas, and highest numbers of properties. The maximum transaction price was SAR 2.437 billion, the maximum property area was 10,249,565 m^2, and the maximum number of properties in a record was 311. These values were investigated as extreme observations, but the IQR procedure did not directly filter transaction price or property area. It filtered only log10(price per m^2). Therefore, extreme values in price or area could remain in the cleaned dataset if their derived price-per-square-meter values fell within the relevant group-specific IQR limits. The notebook does not establish that every extreme record is erroneous or independently validate these records against external sources.

7. Exploratory Data Analysis (EDA)

Important note on dataset stages: The regional, city, property-type, neighborhood, and monthly EDA results in Sections 7.1–7.6 are calculated from the original 201,556-record dataset before outlier filtering. The Power BI dashboard uses the final cleaned dataset of 197,897 records. Therefore, counts and rankings in corresponding visuals may differ; these are results from different processing stages, not necessarily errors.

7.1 Regional Distribution

The dataset covers 13 administrative regions. Riyadh recorded the highest transaction volume (52,770), followed by Makkah (50,950), the Eastern Province (31,156), Qassim (14,332), and Madinah (12,336).

Region	Transactions
Riyadh	52,770
Makkah	50,950

Eastern Province	31,156
Qassim	14,332
Madinah	12,336

7.2 City-Level Distribution

Riyadh (34,060) and Jeddah (31,711) recorded the highest city-level transaction counts, followed by Madinah (10,942) and Makkah (9,683).

City	Transactions
Riyadh	34,060
Jeddah	31,711
Madinah	10,942
Makkah	9,683
Buraidah	6,684
Dammam	5,961
Hail	5,732
Khobar	5,586
Taif	5,496
Hofuf	5,128

7.3 Property Type Distribution

Residential properties represented the largest category, with 182,376 records (approximately 90.48% of the original dataset). Commercial properties accounted for 12,748 records, agricultural properties for 6,431 records, and residential-commercial properties for one record.

Property Type	Transactions
Residential	182,376
Commercial	12,748
Agricultural	6,431
Residential-Commercial	1

7.4 Neighborhood-Level Analysis

Neighborhood transaction counts and price statistics were examined. For both neighborhood rankings, neighborhoods with fewer than 50 transactions were excluded to reduce the influence of very small samples while retaining more locations than the earlier 100-transaction threshold. The median transaction-price ranking shown in Table 6 follows the notebook's neighborhood_stats output and may include general labels such as 'Other'. The separate price-per-square-meter ranking in Table 7 additionally filters names matching the text patterns 'أخرى' or 'خارج حدود الأحياء'. Because this is a text-based filter, spelling variants may remain if they do not match the specified pattern. The dashboard's neighborhood visual is based on the cleaned dataset, so its displayed ranking should be interpreted separately from the original-data EDA table.

Neighborhood	Transactions	Median Price (SAR)
Dammam / West Dammam	97	13,750,000
Jeddah / Al Wadi	57	5,086,000
Diriyah / Other	149	5,000,000
Jeddah / Al Khalidiyah	51	4,370,600
Jeddah / Al Shati	162	4,000,000

7.5 Price per Square Meter

A new feature, Price per Square Meter, was calculated by dividing transaction price by property area. Neighborhoods with at least 50 transactions were considered, while 'Other' and 'Outside Neighborhood Boundaries' categories were excluded from the selected ranking.

Neighborhood	Median Price per m^2 (SAR)
Riyadh / Al Sahafah	12,035.11
Riyadh / Hittin	11,397.21
Makkah / Al Ma'abidah	10,613.46
Riyadh / Al Mughrizat	10,122.87
Riyadh / Al Muruj	9,978.16

7.6 Monthly Transaction Analysis

Transactions were grouped by Gregorian month, and the notebook calculated three measures for each month: transaction count, total transaction value, and average transaction price. Since the dataset covers 2025 only, grouping by month number provides a consistent comparison. February recorded the highest transaction count (22,726) and the highest total transaction value (approximately SAR 26.87 billion). March recorded the highest average transaction price (approximately SAR 1.29 million).

Month	Transactions	Total Value (SAR)	Average Price (SAR)
January	22,319	25.18B	1,128,303
February	22,726	26.87B	1,182,381
March	20,683	26.59B	1,285,589
April	14,264	13.65B	956,925
May	17,456	19.54B	1,119,466
June	10,798	12.16B	1,126,379
July	19,043	17.35B	910,927
August	14,889	14.98B	1,006,039
September	14,888	13.85B	930,116
October	16,255	17.49B	1,075,743
November	12,842	12.92B	1,006,076
December	15,393	18.44B	1,197,902

7.7 Numerical Feature Distributions

Three histograms were added to the updated notebook to visualize transaction price, property area, and price per square meter. The values were transformed using log10 before plotting to make highly skewed ranges easier to inspect and compare.

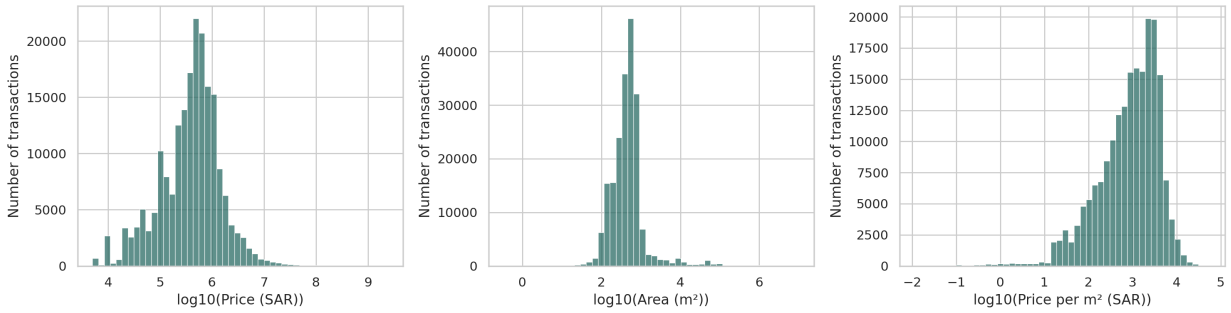


Figure 1. Log10 distributions of transaction price, property area, and price per square meter.

7.8 Outlier Comparison Visualization

A pair of boxplots compares log10 price per square meter by property type before and after outlier removal. The comparison uses the working dataset before filtering and the final dataset after applying the group-based IQR rule.

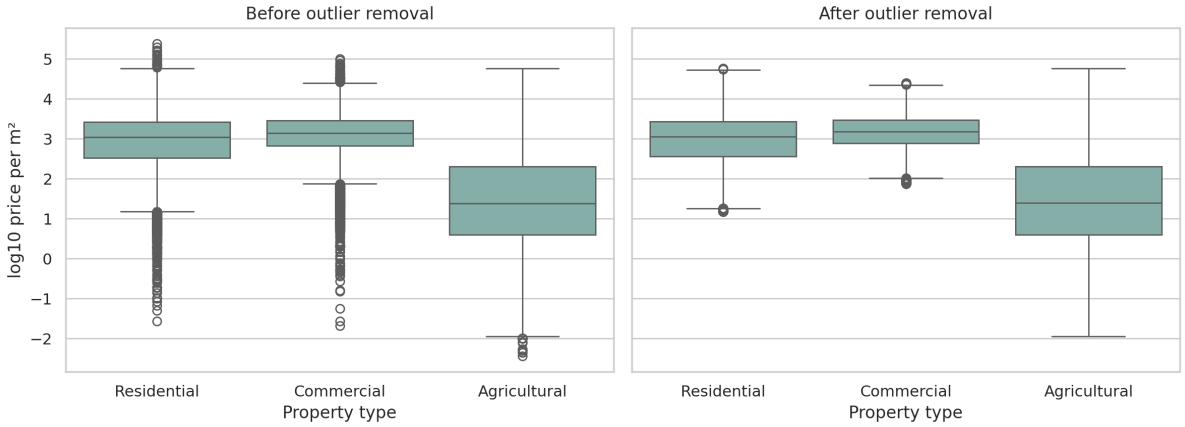


Figure 2. Log10 price-per-square-meter distributions by property type before and after outlier removal.

7.9 Correlation Analysis

A Spearman correlation matrix was calculated using the cleaned dataset. Spearman correlation was selected to assess monotonic relationships without requiring normally distributed variables. The results are shown below.

Variable	Price	Area	Price per m²	Properties count
Price	1.00	0.08	0.76	0.09
Area	0.08	1.00	-0.49	0.12
Price per m²	0.76	-0.49	1.00	0.03
Properties count	0.09	0.12	0.03	1.00

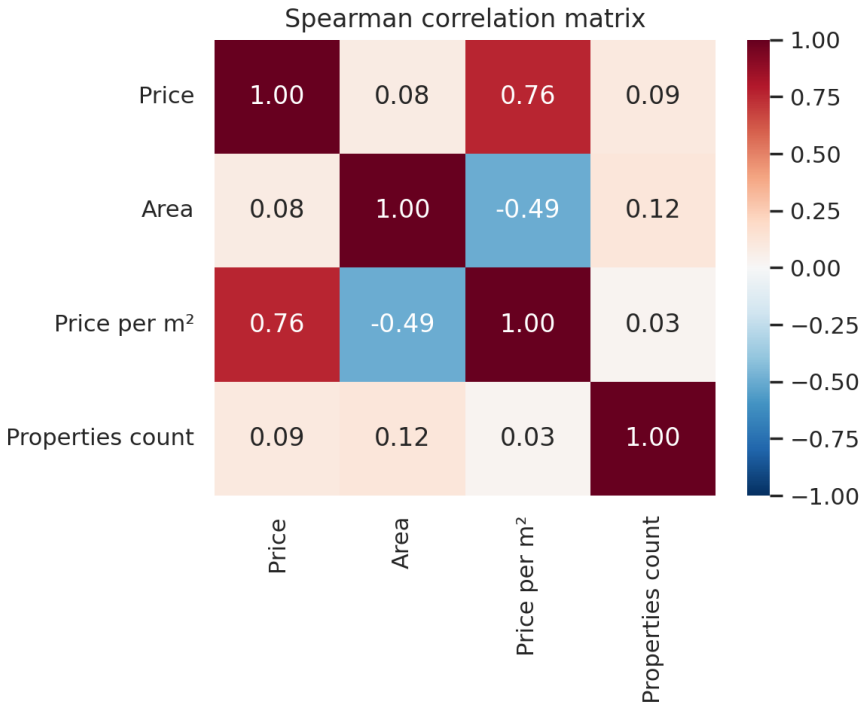


Figure 3. Spearman correlation matrix for the cleaned dataset.

The strongest positive association in the matrix is between transaction price and price per square meter (0.76). Area and price per square meter show a negative association (-0.49). The correlation between price and area is weak (0.08). Price per square meter is calculated from price divided by area, so its correlations with these two variables should be interpreted with that mathematical relationship in mind.

8. Data Cleaning and Outlier Treatment

8.1 Feature Engineering

Price per Square Meter was calculated as transaction price divided by property area. This derived variable was used for outlier detection and neighborhood-level comparisons.

8.2 Excluding the Single-Record Category

The single record classified as Residential-Commercial was excluded before applying the IQR procedure. The working dataset therefore contained 201,555 records. This exclusion is a preprocessing decision based on the category having only one record; it does not by itself establish that the transaction is erroneous.

8.3 IQR-Based Outlier Detection

The IQR method was applied to log10-transformed Price per Square Meter separately within each property type. For each group, Q1 and Q3 were calculated, and IQR was defined as Q3 minus Q1. Observations below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR were excluded. Applying the method separately by property type accounts for differences in the price-per-square-meter distributions of residential, commercial, and agricultural properties. The grouping variable was property type, not administrative region. This choice accounts for broad differences between residential, commercial, and agricultural price-per-square-meter distributions. The notebook did not compare this strategy with region-based or property-type-by-region grouping, so the selected grouping should be understood as the implemented method rather than a demonstrated optimal choice.

8.4 Outlier Removal Results

The IQR procedure removed 3,658 records from the 201,555-record working dataset, leaving 197,897 records. The exact removal rate is approximately 1.8149% (1.81% when rounded to two decimal places), and the retained rate is approximately 98.1851% (98.19% when rounded to two decimal places). The two displayed percentages are rounded independently from their unrounded values.

Measure	Records / Rate
Before outlier removal	201,555
After outlier removal	197,897
Removed	3,658
Removal rate	1.82%
Retained	98.19%

8.5 Removal by Property Type

Commercial properties had the highest within-category removal rate (4.94%), followed by residential properties (1.65%) and agricultural properties (0.28%). Residential properties accounted for the largest number of removed records because they formed the largest category in the working dataset.

Property Type	Before	After	Removed	Removal Rate
Commercial	12,748	12,118	630	4.94%
Agricultural	6,431	6,413	18	0.28%
Residential	182,376	179,366	3,010	1.65%
Total	201,555	197,897	3,658	1.82%

8.6 Impact on Median Price per Square Meter

The median price per square meter increased from SAR 1,045.90 before outlier removal to SAR 1,088.89 after removal, an increase of approximately 4.11%. This indicates that the excluded observations affected the distribution of price per square meter. The change does not imply that every excluded transaction was erroneous; some extreme observations may represent genuine market activity.

Metric	Before Cleaning	After Cleaning
Median Price per m^2 (SAR)	1,045.90	1,088.89

9. Exporting the Cleaned Dataset

The final cleaned dataset was exported as real_estate_cleaned.csv using UTF-8 with BOM encoding. The original dataset was not modified. The exported file contains 197,897 records and is intended for subsequent analysis and Power BI visualization.

10. Key Findings

- The original dataset contains 201,556 real estate transaction records covering 2025.
- No missing values, exact duplicate rows, or non-positive prices and property areas were identified in the reported checks.
- Riyadh and Makkah recorded the highest regional transaction volumes; Riyadh and Jeddah recorded the highest city-level volumes.
- Residential properties represented approximately 90.48% of the original dataset.
- February recorded the highest monthly transaction count, while March recorded the highest average transaction price.
- The updated notebook includes log10 distribution histograms, before-and-after boxplots, and a Spearman correlation heatmap.
- The IQR procedure removed 3,658 records (1.81%) from the 201,555-record working dataset.
- Commercial properties had the highest within-category removal rate (4.94%).
- The median price per square meter increased by approximately 4.11% after outlier removal.

11. Conclusion

The project applied data inspection, descriptive statistics, exploratory analysis, feature engineering, visualization, and outlier detection to Saudi real estate transaction data for 2025. The updated notebook now includes feature-distribution plots, a before-and-after outlier comparison, and Spearman correlation analysis in addition to geographical, neighborhood, and monthly analyses. A group-based IQR method on log10 price per square meter produced a cleaned dataset of 197,897 records. This dataset provides a prepared basis for further analysis and interactive reporting in Power BI.

12. Scope and Interpretation Notes

- The notebook checks exact-row duplicates. A separate validation of the transaction reference number in the source dataset found 201,556 non-missing unique references and zero duplicate reference numbers.

Counts in the notebook's original-data EDA tables (201,556 records) are not expected to match all corresponding Power BI counts, because the dashboard is based on the cleaned 197,897-record dataset.

- The correlation between price per square meter and price or area is partly influenced by the formula used to calculate price per square meter.
- The single Residential-Commercial record was excluded from the working dataset because it formed a one-record category; the notebook does not establish that it was invalid.
- The EDA distribution plots and correlation matrix use the final cleaned dataset, while the before-and-after boxplots compare the working and cleaned datasets.

All dashboard visuals and KPI cards are based on the final cleaned dataset (197,897 records), unless a visual is explicitly described otherwise. The original-data EDA counts in Section 7 are retained to document the notebook's initial exploration.

13. Power BI Dashboard

An interactive Power BI dashboard was developed to communicate the main findings from the cleaned 2025 real estate transactions dataset. The dashboard contains two pages: an overview page and a price analysis page. Both pages display the same five high-level indicators based on the cleaned dataset.

13.1 Dashboard Filters

The report pages include slicers for region, city, neighborhood, property type, and time period. These filters are intended to allow users to focus the visuals on selected geographical areas, property categories, and periods.

13.2 Page 1: General Overview

The overview page presents the overall transaction activity and geographical distribution. It includes KPI cards for total transactions, median transaction price, total transaction value, median price per square meter, and median property area. The supporting visuals show the highest neighborhoods by median transaction price, transaction counts by city, a geographical map, property-type distribution, monthly transaction value and median price, and transaction counts by quarter.

- The cleaned dataset contains 197,897 transactions.
- The median transaction price is SAR 450,000.
- The total transaction value is approximately SAR 211.299 billion.
- The median price per square meter is approximately SAR 1,089.
- The median property area is 456 m^2.

The city ranking highlights Riyadh and Jeddah as the two cities with the highest transaction counts in the dashboard. The quarterly visual provides a high-level comparison of transaction activity across the four quarters of 2025.

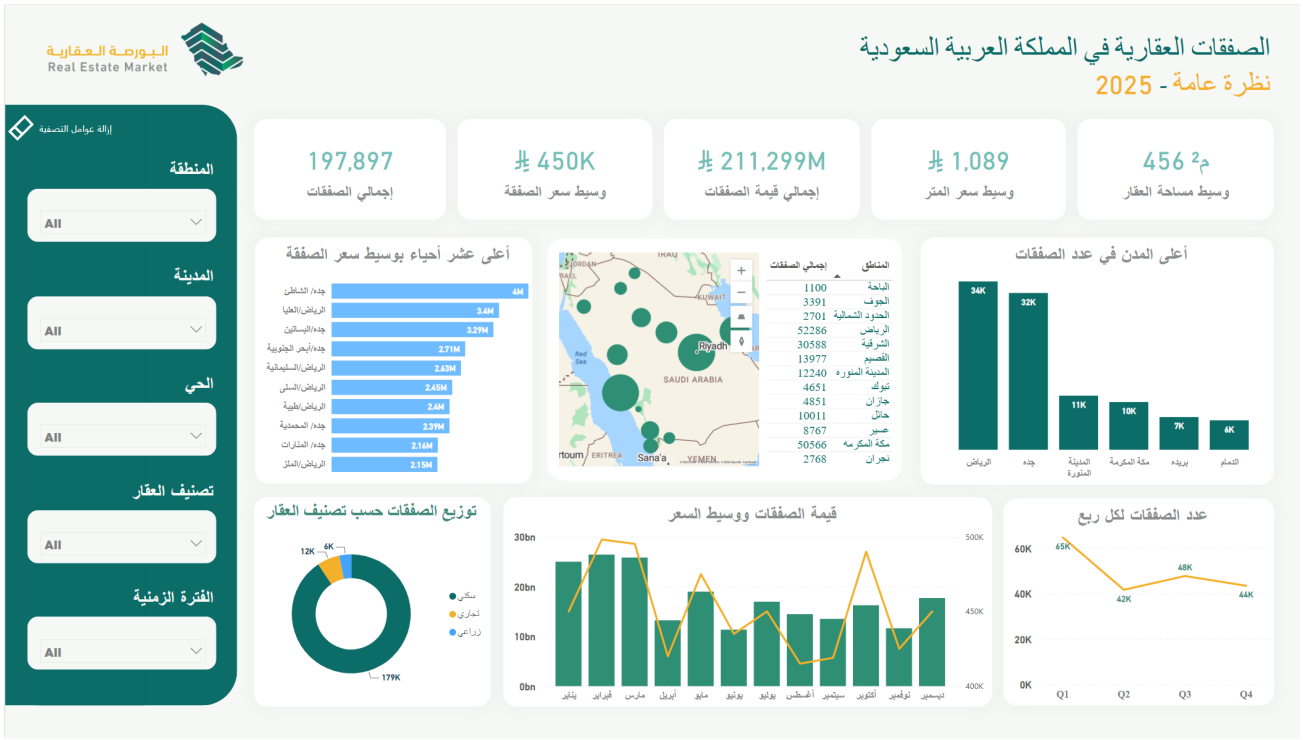


Figure 4. Updated Power BI dashboard — General Overview page.

13.3 Page 2: Price Analysis

The price analysis page focuses on price-per-square-meter patterns and the relationship between property area and transaction price. It includes median price per square meter by property type, median price per square meter by property-area category, transaction distribution across price-per-square-meter bands, regional median price per square meter, and a scatter plot comparing area and transaction price.

The page supports comparisons between property categories and area bands. The scatter plot helps inspect the spread of transaction prices across different property sizes; the logarithmic axes make the wide range of values easier to display.

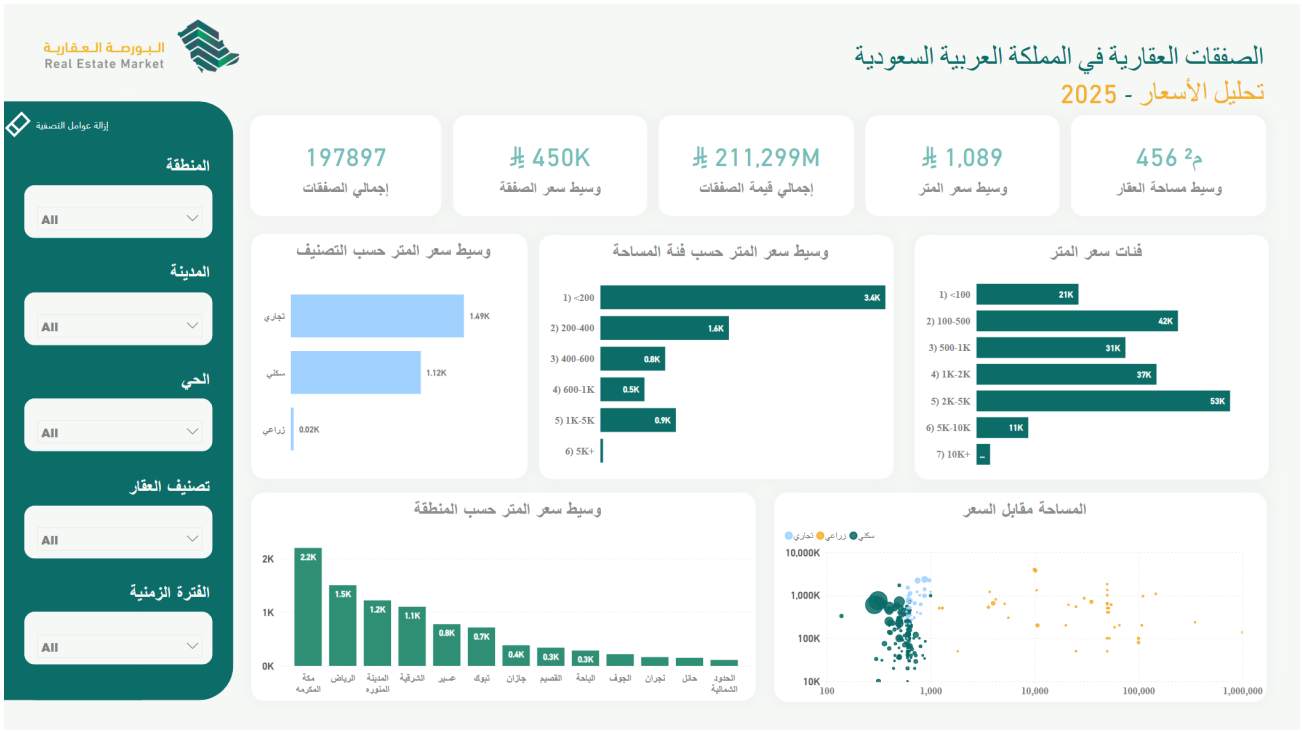


Figure 5. Updated Power BI dashboard — Price Analysis page.

14. Reconciliation of Notebook and Dashboard Metrics

The notebook contains two analysis stages. First, the EDA tables and descriptive counts in Sections 4–7 use the original dataset of 201,556 records. Second, the notebook excludes the single Residential-Commercial record and applies the group-based IQR rule to log10 price per square meter, producing a cleaned dataset of 197,897 records. The Power BI dashboard uses this cleaned dataset. As a result, regional, city, property-type, neighborhood, and other transaction counts may differ between the original-data EDA tables and dashboard visuals.

Metric	Original dataset	Cleaned dataset / Dashboard
Number of records	201,556	197,897
Median transaction price	SAR 445,852.50	SAR 450,000
Median property area	465.65 m^2	456 m^2
Median price per m^2	Not reported in original descriptive table	SAR 1,088.89 (shown as 1,089)
Total transaction value	Not reported in original descriptive table	SAR 211,299,000,735.89 (shown as 211,299M)

The dashboard's median property area of 456 m^2 is consistent with the cleaned dataset. The value of 465.65 m^2 is the median area in the original, unfiltered dataset. Both values are valid for their respective dataset stages.

The dashboard abbreviates some KPI values for readability. For example, 450K represents SAR 450,000 and 1,089 represents approximately SAR 1,088.89 per square meter. The total transaction value is approximately SAR 211.299 billion; the dashboard label '211,299M' is mathematically interpretable but less familiar to readers. A clearer display is 'SAR 211.3

billion'. Regional and city counts shown in the dashboard should likewise be read as cleaned-dataset counts, while the corresponding tables in Sections 7.1–7.2 report original-dataset counts from the notebook.

15. Power BI DAX Calculations

This section documents the DAX calculated columns and measures created in Power B). The calculations are defined on the real_estate_cleaned table and support the dashboard KPIs, time analysis, and categorical comparisons.

15.1 Calculated Columns

Calculated columns are evaluated row by row and stored in the model. They provide labels and grouping fields used by slicers and visuals.

Area Category

Groups properties into area bands for comparing transaction patterns by property size.

```
فئة المساحة =
SWITCH(
    TRUE(),
    real_estate_cleaned200) <1, "200] < المساحة[",
    real_estate_cleaned400-200) 2, "400] < المساحة[",
    real_estate_cleaned600-400) 3, "600] < المساحة[",
    real_estate_cleaned1-600) 4, "1000] < المساحة[K",
    real_estate_cleaned1) 5, "5000] < المساحة[K-5K",
    "6) 5K+"
)
```

Price per Square Meter Category

Creates price-per-square-meter bands used in the price distribution visual.

```
فئة سعر المتر =
SWITCH(
    TRUE(),
    real_estate_cleaned100) <1, "100] < سعر المتر[",
    real_estate_cleaned500-100) 2, "500] < سعر المتر[",
    real_estate_cleaned1-500) 3, "1000] < سعر المتر[K",
    real_estate_cleaned1) 4, "2000] < سعر المتر[K-2K",
    real_estate_cleaned2) 5, "5000] < سعر المتر[K-5K",
    real_estate_cleaned5) 6, "10000] < سعر المتر[K-10K",
    "7) 10K+"
)
```

Month Number

Extracts the month number (1–12) from the Gregorian transaction date.

```
رقم الشهر =
MONTH('real_estate_cleaned'[تاريخ الصفقة ميلادي])
```

Quarter Number

Extracts the quarter number (1–4) from the transaction date.

```
رقم الربع =
QUARTER('real_estate_cleaned'[تاريخ الصفقة ميلادي])
```

Clean Region Name

Removes the prefix 'منطقة' and trims extra spaces to standardize region labels.

```
المناطق =
TRIM(
    SUBSTITUTE(
        'real_estate_cleaned'[المناطق],
```

```
        "منطقة ",
        ""
    )
)
```

Quarter Label

Creates readable quarter labels such as Q1, Q2, Q3, and Q4.

```
الربع =
"Q" & QUARTER('real_estate_cleaned'[تاريخ الصفقة ميلادي])
```

Month Name

Maps month numbers to Arabic month names for chart labels.

```
اسم الشهر =
SWITCH(
    'real_estate_cleaned'[رقم الشهر],
    1, "يناير",
    2, "فبراير",
    3, "مارس",
    4, "أبريل",
    5, "مايو",
    6, "يونيو",
    7, "يوليو",
    8, "أغسطس",
    9, "سبتمبر",
    10, "أكتوبر",
    11, "نوفمبر",
    12, "ديسمبر"
)
```

15.2 Measures

Measures are evaluated dynamically according to the filter context in the report. They are used in KPI cards and visual aggregations.

Transaction Count (عدد الصفقات)

Counts the rows in the cleaned transactions table.

```
عدد الصفقات =
COUNTROWS(real_estate_cleaned)
```

Total Transactions (إجمالي الصفقات)

Counts nonblank transaction reference numbers. This equals the row count when every row has a reference number.

```
إجمالي الصفقات =
COUNT('real_estate_cleaned'[الرقم المرجعي للصفقة])
```

Total Transaction Value (إجمالي قيمة الصفقات)

Sums transaction prices to calculate total transaction value.

```
إجمالي قيمة الصفقات =
SUM('real_estate_cleaned'[السعر])
```

Total Property Area (إجمالي مساحة العقارات)

Sums property areas in the current filter context.

```
إجمالي مساحة العقارات =
SUM('real_estate_cleaned'[المساحة])
```

Median Transaction Price (وسيط سعر الصفقة)

Returns the median transaction price, reducing sensitivity to very high transaction values.

```
وسيط سعر الصفقة =
MEDIAN('real_estate_cleaned'[السعر])
```

Median Price per Square Meter (وسيط سعر المتر)

Returns the median price per square meter.

```
وسيط سعر المتر =
MEDIAN('real_estate_cleaned'[سعر المتر])
```

Median Property Area (وسيط مساحة العقار)

Returns the median property area.

```
وسيط مساحة العقار =
MEDIAN('real_estate_cleaned'[المساحة])
```

Average Property Area (متوسط مساحة العقار)

Calculates the arithmetic mean of property area.

```
متوسط مساحة العقار =
AVERAGE('real_estate_cleaned'[المساحة])
```

Average Price per Square Meter (متوسط سعر المتر)

Calculates the arithmetic mean of price per square meter.

```
متوسط سعر المتر =
AVERAGE('real_estate_cleaned'[سعر المتر])
```

Average Transaction Price (متوسط سعر الصفقة)

Calculates the arithmetic mean of transaction price.

```
متوسط سعر الصفقة =
AVERAGE('real_estate_cleaned'[السعر])
```

Median Property Area for Cities (30+ Transactions) (وسيط المساحة للمدن)

Returns the median area only when the current city/filter context contains at least 30 transactions.

```
وسيط المساحة للمدن =
IF(
    30] >= إجمالي الصفقات[,
    MEDIAN(real_estate_cleaned[المساحة])
)
```

Median Price per Square Meter for Cities (30+ Transactions) (وسيط سعر المتر للمدن)

Returns the median price per square meter only when at least 30 transactions are present in the current context.

```
وسيط سعر المتر للمدن =
IF(
    30] >= إجمالي الصفقات[,
    MEDIAN(real_estate_cleaned[سعر المتر])
)
```

Median Transaction Price (100+ Transactions) (وسيط السعر 100+)

Returns the median transaction price only when the current context contains at least 100 transactions.

```
وسيط السعر 100+ =
IF(
    100] >= إجمالي الصفقات[,
    MEDIAN(real_estate_cleaned[السعر])
)
```

15.3 How the DAX Supports the Dashboard

- The transaction-count and total-value measures support the headline KPI cards.
- Median measures support the transaction-price, price-per-square-meter, and property-area KPIs.
- Month number, month name, quarter number, and quarter label support monthly and quarterly visuals.
- Area Category and Price per Square Meter Category create consistent bins for distribution comparisons.
- The cleaned region label supports clearer regional grouping and map/chart labels.
- The conditional median measures apply minimum transaction-count rules (30 or 100) before returning a median, helping avoid displaying selected statistics for very small groups.

15.4 DAX Interpretation Notes

- COUNTROWS(real_estate_cleaned) counts all rows, whereas COUNT on the transaction reference column counts only nonblank reference values. The two measures should agree when reference numbers are complete.
- The 30-transaction and 100-transaction thresholds are explicit business/reporting rules in the DAX measures. They are not statistical guarantees of representativeness.
- The month-name column is text. For chronological ordering in visuals, sort the month-name column by the numeric Month Number column.
- The area and price-per-square-meter categories use ordered text labels (1), (2), etc. If sorting is not chronological/numeric, use a corresponding numeric sort column.

16. Final Project Summary

The completed project combines Python-based data inspection, preprocessing, exploratory data analysis, statistical analysis, and visualizations with an interactive Power BI dashboard. The notebook documents the data-quality checks, descriptive statistics, geographical and monthly patterns, price-per-square-meter feature engineering, IQR-based outlier treatment, and the impact of filtering. The Power BI report presents the main results through an overview page and a dedicated price analysis page.

The final cleaned dataset contains 197,897 records and is the basis for the Power BI dashboard. The notebook's initial EDA tables retain statistics from the original 201,556-record dataset, and the report explicitly distinguishes those results from cleaned-dashboard metrics. The original dataset remains unchanged. Together, the notebook, cleaned CSV output, technical report, and Power BI dashboard provide a documented analysis workflow for the 2025 real estate transaction data.